

What Makes Deep Research?

A Dive into AI Agents

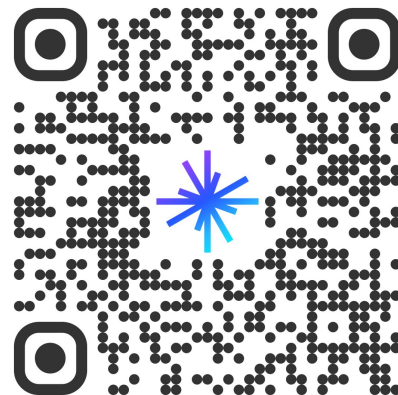


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- 03 What makes it "tick"?**
- 03 Where to from here?**

01

New Kid on the Block: Deep Research

Share



Activity

15 sources



Experience Level: Intermediate

Snowboard Type: Freestyle

Height: 180cm

Weight: 89kg

Shoe Size: 11

Budget: 500-700

Great! I will research freestyle snowboards suitable for an intermediate rider with your specifications and within your budget range of \$500-\$700. I'll focus on options that balance performance and durability for tricks and jumps. I'll update you soon with the best options.

Researched for 8m32s · 15 sources

Snowboards for Intermediate Freestyle Riders

For an intermediate rider (~180 cm, 89 kg, US 11 boots) seeking a freestyle board, here are some top options (\$500-\$700) that excel at park tricks/jumps, offer durability, and accommodate your size:

Citations

Evo

Ride TwinPig Snowboard: A Game Changer for Snowboarders
September 29, 2024 — Snowboarders were ecstatic to expe...

The Ride TwinPig brought an exhilarating surprise to snowboarders this weekend, reminiscent of the excitement a...

Snowboard Review Hub

The GNU Riders Choice is the Best All-Mountain Snowboard for Every Adventure.

The Riders Choice is a versatile, high-performance snowboard perfect for all conditions. Priced at \$499...

Burton

GNU Riders Choice: Snowboarders Praise Its Flexibility and Control...

Exceptional control and flexible design, especially after its impressive performance in various snowboarding competitio...

GNU

GNU Riders Choice: Crafted for Your Ultimate Ride.

Experience unparalleled performance and stability with the latest GNU snowboard.

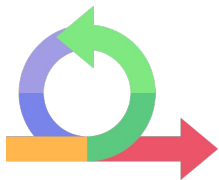
All Sources

Evo Snowboard Review Hub Burton REI

What was new about Deep Research?

- Wasn't first in this category
 - Google's Deep Research released in Dec, 2024
- Focus on end-to-end training of chain-of-thought reasoning with RL?
- Something else?

Research Agents

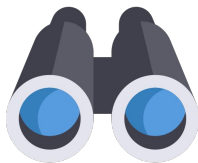


Iteration

“...learned to plan and execute a multi-step trajectory...”

“...backtracking and reacting to real-time information...”

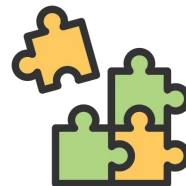
“...pivoting as needed in reaction to information it encounters...”



Search

“...trained using end-to-end reinforcement learning on hard browsing and reasoning tasks across a range of domains...”

“...optimized for web browsing and data analysis...”



Reasoning

“...fine-tuned on the upcoming OpenAI o3 reasoning model...”

““...leverages reasoning to search, interpret, and analyze massive amounts of text...””

02

Code Walkthrough Zilliz's DeepSearcher

About Milvus

Milvus is an open-source vector database for GenAI projects. pip install on your laptop, plug into popular AI dev tools, and push to production with a single line of code.



33K+

GitHub Stars



2.7K+

Forks



66M+

Docker Pulls



400+

Contributors



Easy Setup

Pip-install to start coding in a notebook within seconds

Reusable Code

Write once, and deploy with one line of code into the production environment

Integration

Plug into OpenAI, Langchain, LlmIndex, and many more

Feature-rich

Dense & sparse embeddings, filtering, reranking and beyond

Milvus Users

accenture

airbnb

AT&T



BOSCH

Chegg

CISCO

CISION

COMPASS

Deloitte.

ebay

FARFETCH

Grab

IKEA

Inflection

intuit.

Microsoft

new relic.

nVIDIA

OMERS

Otter.ai

PayPal

paloalto
NETWORKS

POSHMARK

ROBLOX

salesforce

Shell

shutterstock

T

TREND
MICRO

Walmart



ZipRecruiter

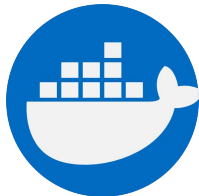
zomato

Deployment Options



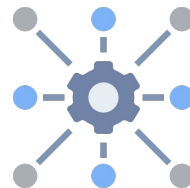
Milvus Lite

- Locally hosted
- Suitable for prototyping and demos



Milvus Standalone

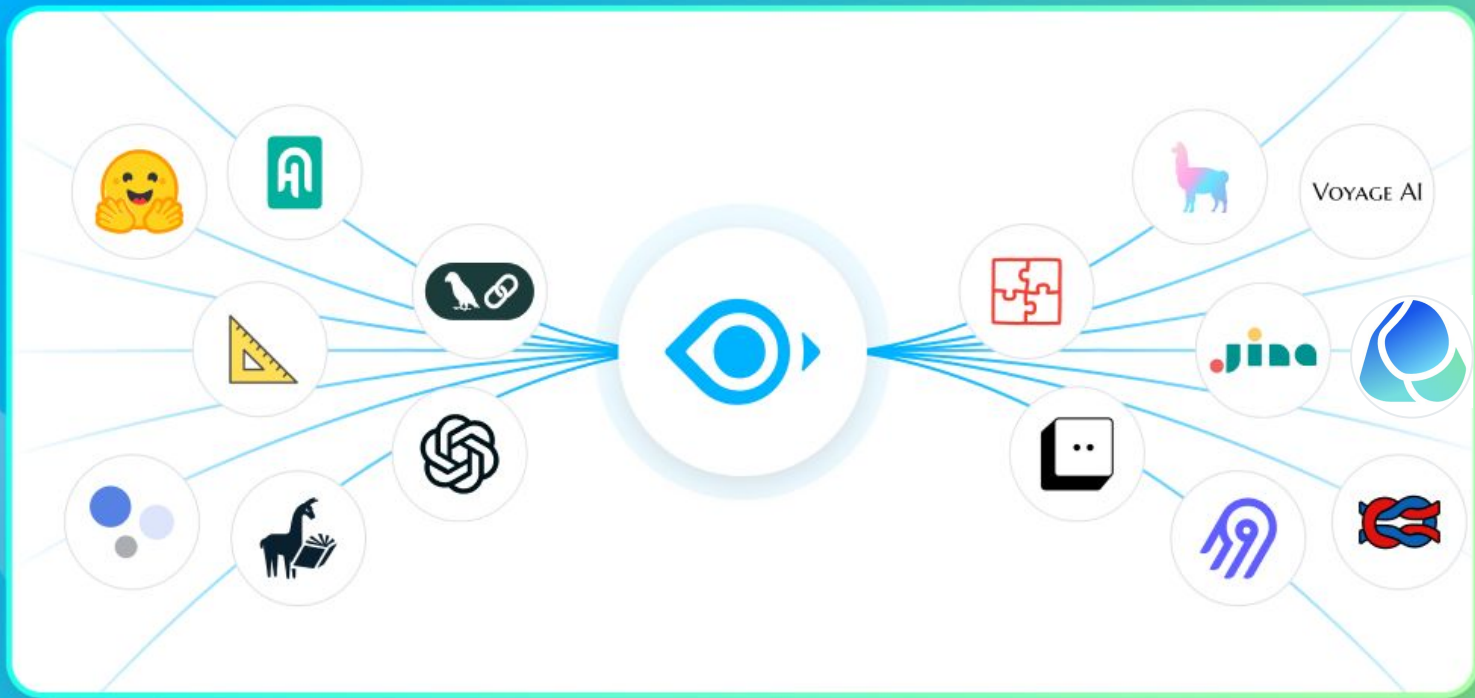
- Single remote/local server
- "Medium" scale
- Simplified setup, maintenance, etc. compared to cluster



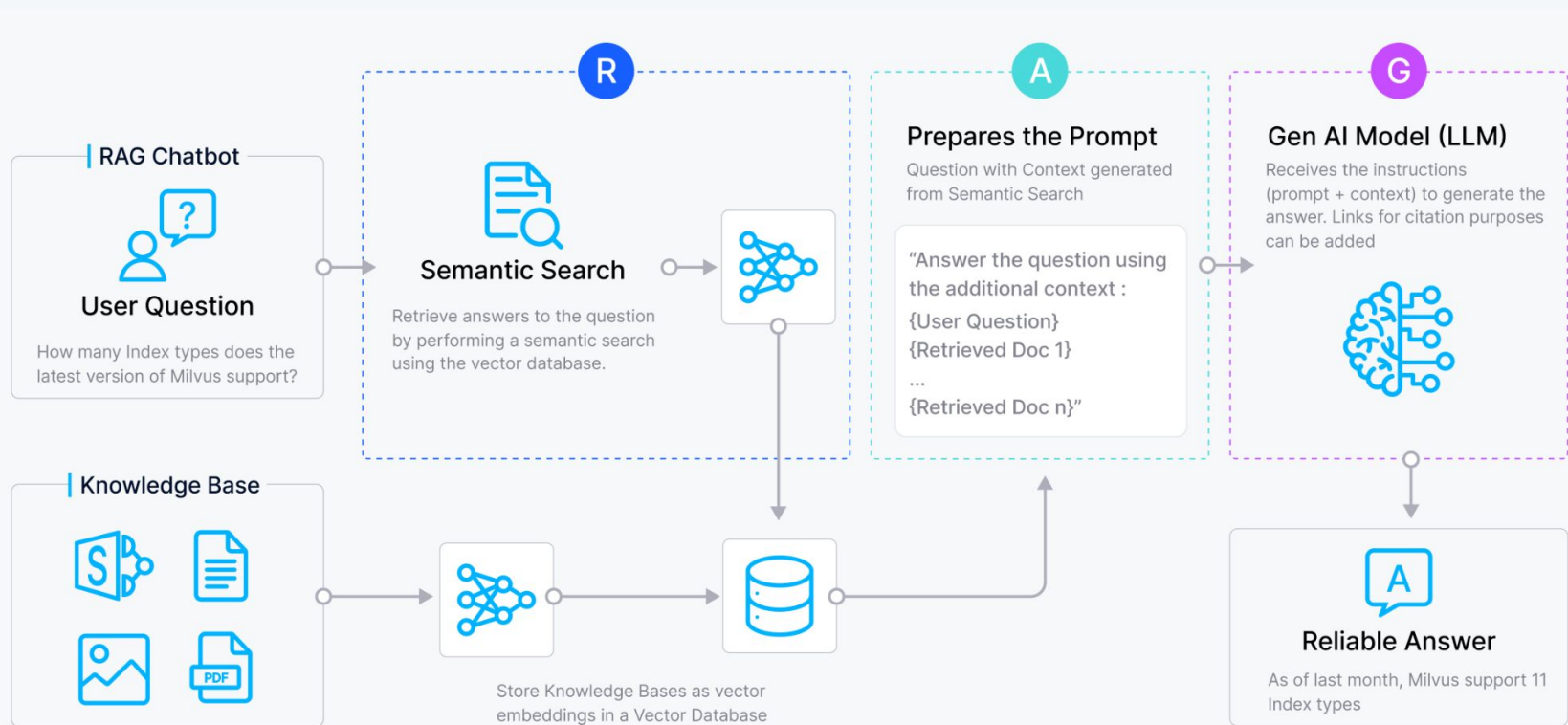
Milvus Cluster

- Distributed system
- Many different types of nodes
- Scales to 100s of billions of vectors

Seamless integration with all popular AI toolkits

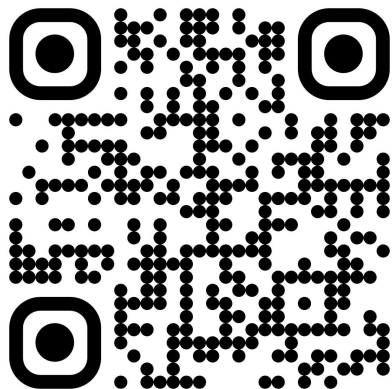


Retrieval-Augmented Generation

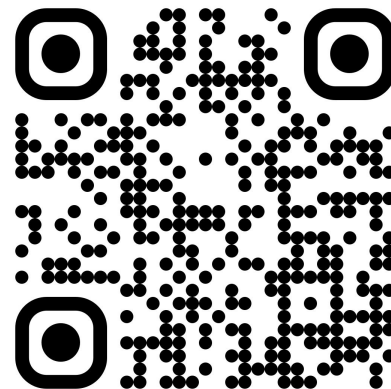




milvus

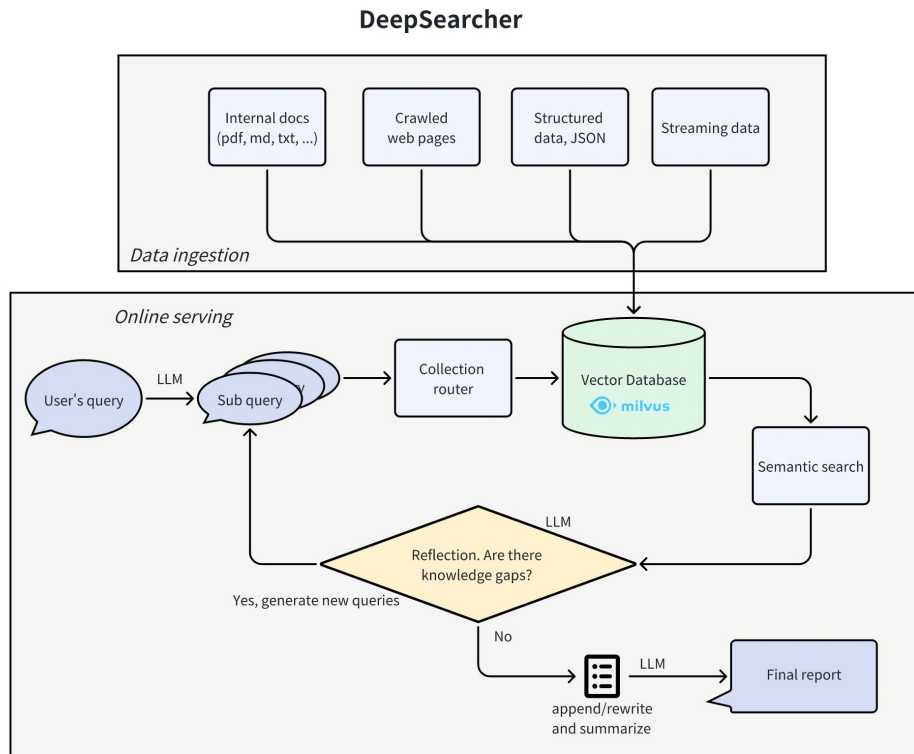


github.com/milvus-io/milvus



zilliz.com/learn/generative-ai

Code Walkthrough



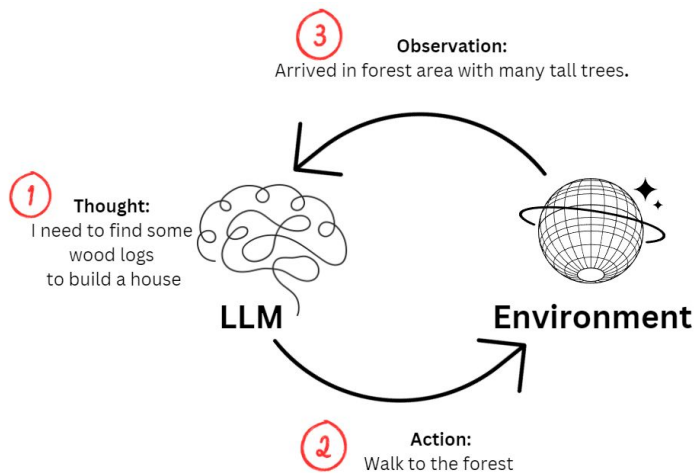
03

What makes it "tick"?

Agents — Tool Usage

- Ability to call functions
- Equivalently, write and run code (in sandbox)
- Search web
- Use of structured output / constrained sampling

Agents — Cognitive Models



(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: `Search[Apple Remote]`

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the `Front Row media center` program ...

Thought 2: Apple Remote was originally designed to control the `Front Row media center` program. I need to search `Front Row` next and find what other device can control it.

Act 2: `Search[Front Row]`

Obs 2: Could not find `[Front Row]`. Similar: `['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]`

Thought 3: `Front Row` is not found. I need to search `Front Row (software)` .

Act 3: `Search[Front Row (software)]`

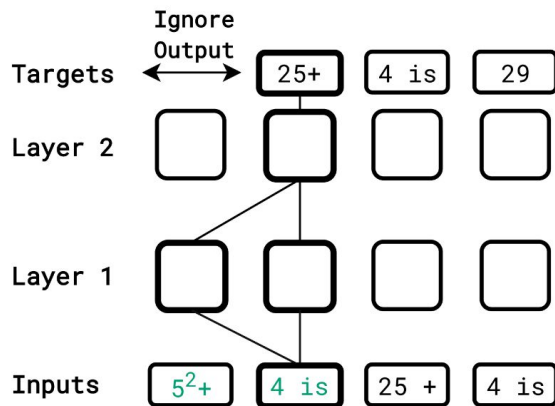
Obs 3: `Front Row` is a discontinued media center software ...

Thought 4: `Front Row (software)` is controlled by an `Apple Remote` or the `keyboard function keys`. So the answer is `keyboard function keys`.

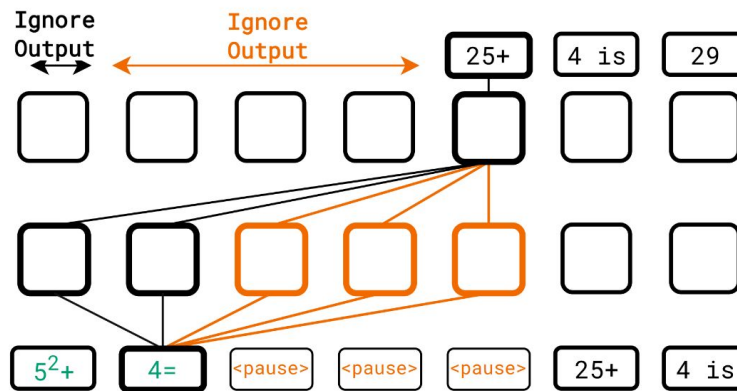
Act 4: `Finish[keyboard function keys]`



Reasoning — Conditional Computation

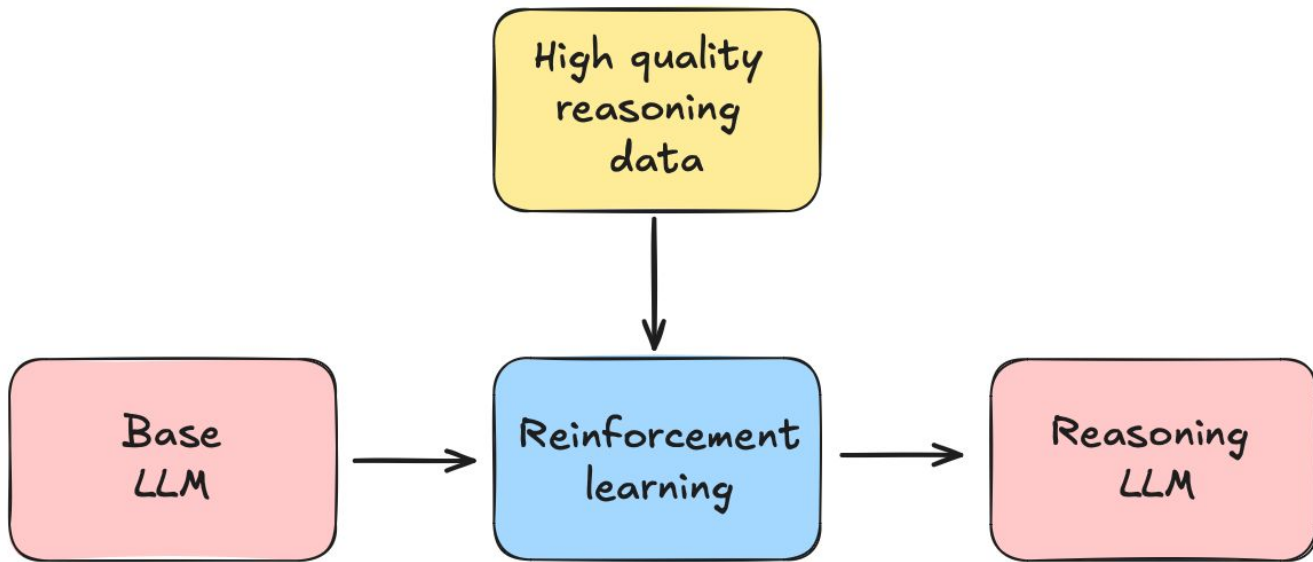


(a) Standard inference and finetuning

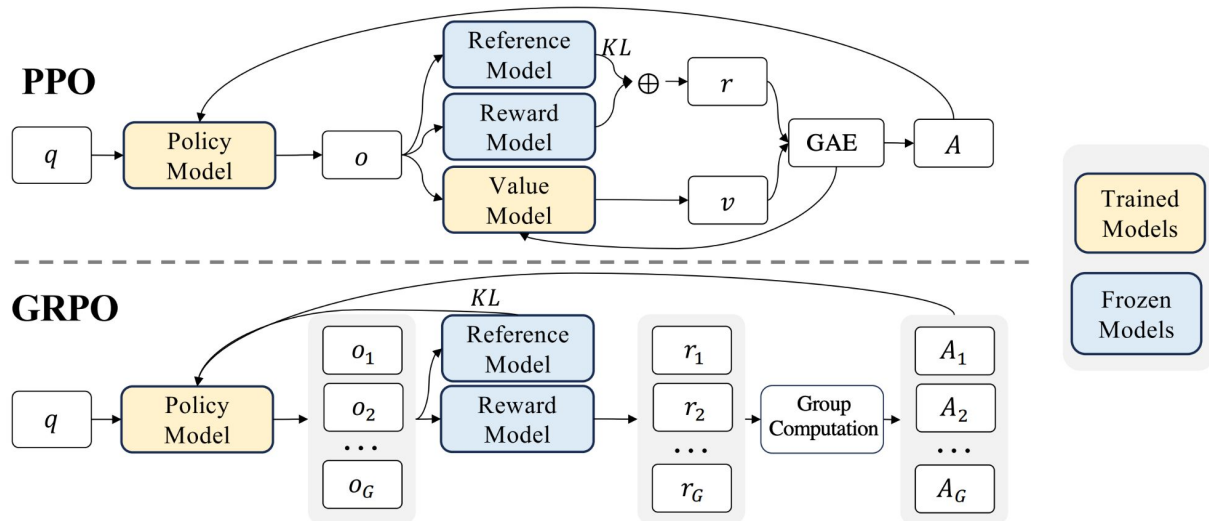


(b) Pause-inference and finetuning

Reasoning — GRPO Training



Reasoning — GRPO Training



04

Where to from here?

Challenges

- Cost
- Hallucinations (factual errors)
- Reasoning errors
- Open-source reasoning trace datasets

Where to from here?

- Cost
 - Special hardware (SambaNova, Cerebras)
 - Continuous CoT
- Barriers to entry
 - Milvus
 - ["I Built a Deep Research with Open Source—and So Can You!"](#)
 - ["Introducing DeepSearcher: A Local Open Source Deep Research"](#)
 - HuggingFace
 - ["Open-R1: a fully open reproduction of DeepSeek-R1"](#)
 - ["Open-source DeepResearch – Freeing our search agents"](#)

LET'S STAY CONNECTED!

Stefan Webb

Developer Advocate, Zilliz



<https://milvus.io/discord>



<https://github.com/milvus-io/milvus>



<https://x.com/milvusio>



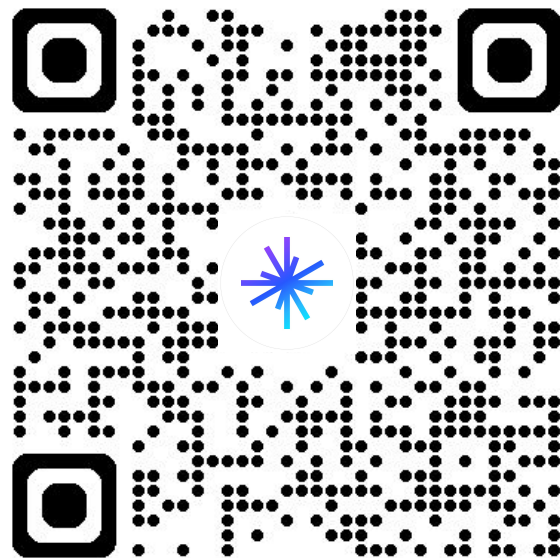
<https://www.linkedin.com/company/the-milvus-project>



Unstructured Data Podcast

Latest Episodes

- Inside the AI Revolution
- Prompt, Score, Repeat: Principled RAG and Agent Design





Workshop with Milvus and OpenAI

Join us for a hands-on session with
OpenAI to learn about Agents!



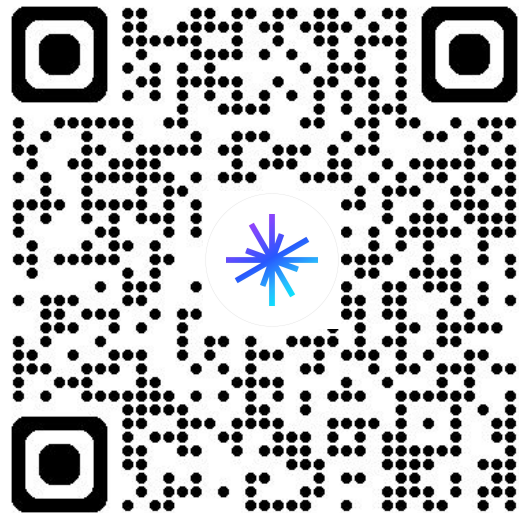
March 20, 2025



5:30 - 8:30 PM



Palo Alto





Book a free 1:1 session to get help with your production deployment
meetings.hubspot.com/chloe-williams1/milvus-office-hours

THANK YOU

